

Domain-Generalizable Phishing URL Detection under Realistic Evaluation Splits

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Abstract

Machine-learning phishing detectors often report very high accuracy when URL datasets are split randomly. However, a real detector must identify phishing URLs from domains it has never seen before. We study phishing URL detection using realistic evaluation splits. In the main test, every registered domain (effective top-level domain plus one, or eTLD+1) in the test set is excluded from training. We compare random, host-disjoint, eTLD+1-disjoint, and cross-source splits to measure how random splits can make results look better than they really are.

1 Introduction and Motivation

Phishing pages pretend to be trusted websites so they can steal passwords and financial information. The Anti-Phishing Working Group recorded 3.8 million attacks in 2025 [2]. A system can inspect a URL before a user opens it, so accurate detection could protect browsers, email systems, and messaging apps. Previous work reports strong machine-learning results [3], but a useful detector must find attacks from new domains without blocking too many safe URLs.

1.1 Motivating Example

Consider a message asking a user to verify a PayPal account at <https://paypal.login-fraud.top>. The URL contains the trusted name “PayPal,” but its registered domain is `login-fraud.top`. If the system marks this URL as phishing before it opens, it may stop the user from entering a password, even though the domain was not in the training data.

2 Gap and Project Objective

Random splits can place similar URLs in both training and test sets. A model may then remember domains instead of learning general phishing patterns. In earlier work, performance dropped by 10–32% when models were tested on a different dataset [4]. A 2026 study used separate hosts but still found 1,387 shared eTLD+1 domains in training and testing [1]. This means that subdomains of the same registered domain can appear on both sides of the split.

We will group URLs by eTLD+1 and compare random-record, host-disjoint, eTLD+1-disjoint, and cross-source tests. Experiments will use the PhiUSIIL dataset of 235,795 websites [3] and a second public URL dataset. We will measure recall, precision, macro-F1, Matthews correlation, false-positive rate, and calibration. The project will provide a repeatable splitting method and show how much random splitting may overestimate real-world performance.

3 Formal Problem Definition

Let $\mathcal{U}$ be the set of URL strings, $\mathcal{Y} = \{0, 1\}$, and $r : \mathcal{U} \rightarrow \mathcal{R}$ map a URL to its registrable domain (eTLD+1). Label $y = 1$ denotes phishing and $y = 0$ denotes legitimate. Given a labeled training corpus $D_{\text{tr}} = \{(u_i, y_i)\}_{i=1}^n \subset \mathcal{U} \times \mathcal{Y}$, the task is to learn a function

$$f : \mathcal{U} \rightarrow \mathcal{Y}$$

that predicts the label of each test URL u . The registered domain of every test URL must be absent from the training set:

$$r(u) \notin \{r(u_i) : (u_i, y_i) \in D_{\text{tr}}\}.$$

For cross-source testing, the test URLs also come from a different dataset. The output is one binary label for each URL. This definition does not require any specific model or set of features.

3.1 Example Instance

For $u = \text{https://paypal.login-fraud.top}$, suppose `login-fraud.top` is absent from training. The expected output is $f(u) = 1$ (phishing).

4 Team and Collaboration

Prabhnoor leads data-source checks and split validation. Divraj leads model testing and makes sure experiments can be repeated. Dilraj leads dataset integration, experiment tracking, repository maintenance, and error analysis. The team combines data engineering, machine learning, and cybersecurity skills. All members help with problem definition, literature review, implementation, experiments, analysis, and writing. Repository: <https://github.com/prabhnoor/domain-generalizable-phishing-url-detection>.

References

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